

Comparison of while, do-while, and for Loops

Feature	while Loop	do-while Loop	for Loop
Syntax	<pre>while (condition) { // body }</pre>	<pre>do { // body } while (condition);</pre>	<pre>for (initialization; condition; update) { // body }</pre>
Condition Evaluation	Condition is checked before entering the loop.	Condition is checked after entering the loop.	Condition is checked before each iteration.
Loop Execution	Executes 0 or more times (if condition is false initially, the loop doesn't execute).	Executes 1 or more times (the loop executes at least once).	Executes 0 or more times (depends on the condition).
Ideal Use Case	When the number of iterations is not known and the loop might not execute at all.	When the loop should execute at least once, regardless of the condition.	When the number of iterations is known beforehand or can be easily controlled.
Loop Initialization	Initialization must be done before the loop starts (outside the loop).	Initialization is done outside the loop, similar to <code>while</code> .	Initialization is part of the loop statement, making it concise and compact.